



Predicting College Basketball Spreads using Schedule-Adjusted and Context-Aware Features

Yunus Mouline¹ and Shane Faberman¹

¹Department of Statistics and Operations Research, Sports Analytics Intelligence Lab, University of North Carolina, Chapel Hill, NC, 27599, USA



INTRODUCTION



Although team strength is often inferred from record or reputation, predicting college basketball game margins requires a more rigorous understanding of team quality. This project builds a pregame spread prediction framework using schedule-adjusted efficiency, play style, roster variation, and contextual features. By combining multi-season public game data with predictive modeling, the goal is to better explain and forecast margin outcomes.

METHODS

A rolling-window framework was designed so that the model is trained only on information available before each game, preventing information leakage.

DATA COLLECTION

Compiled multi-season public college basketball data from play-by-play and box score sources, creating a game-level dataset with team performance, schedule, and roster context.

FEATURE ENGINEERING

Constructed rolling-window features capturing context, schedule-adjusted efficiency, play style, and lineup availability so each game is predicted only by pre-game information.

MODELING

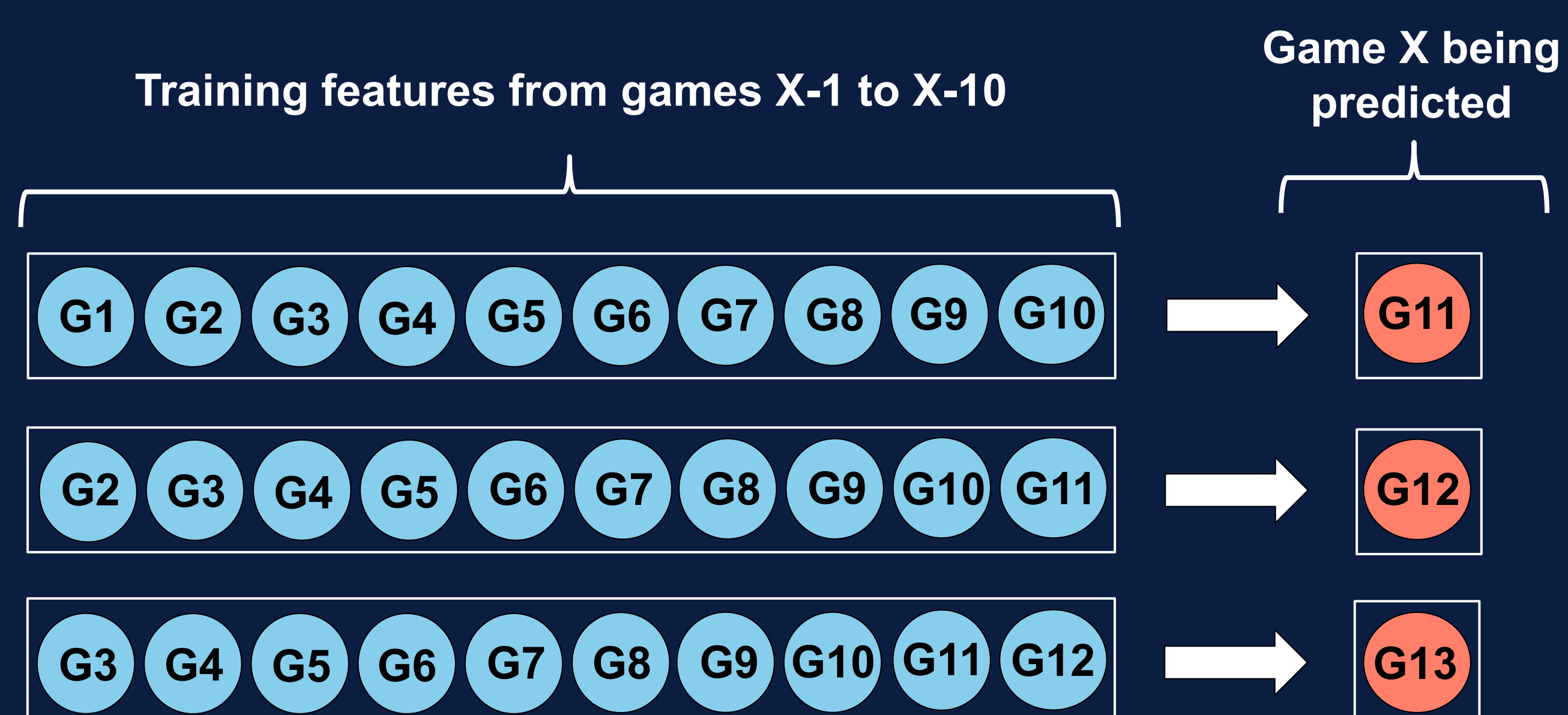
Tested multiple predictive models, including XGBoost, CatBoost, and Ridge Regression, to estimate game margin and compare how different structures interacted with the feature set.

FUTURE PREDICTION

Generated future-game rows using only available pregame inputs, allowing the framework to estimate expected margin for upcoming matchups in a realistic forecasting setting.

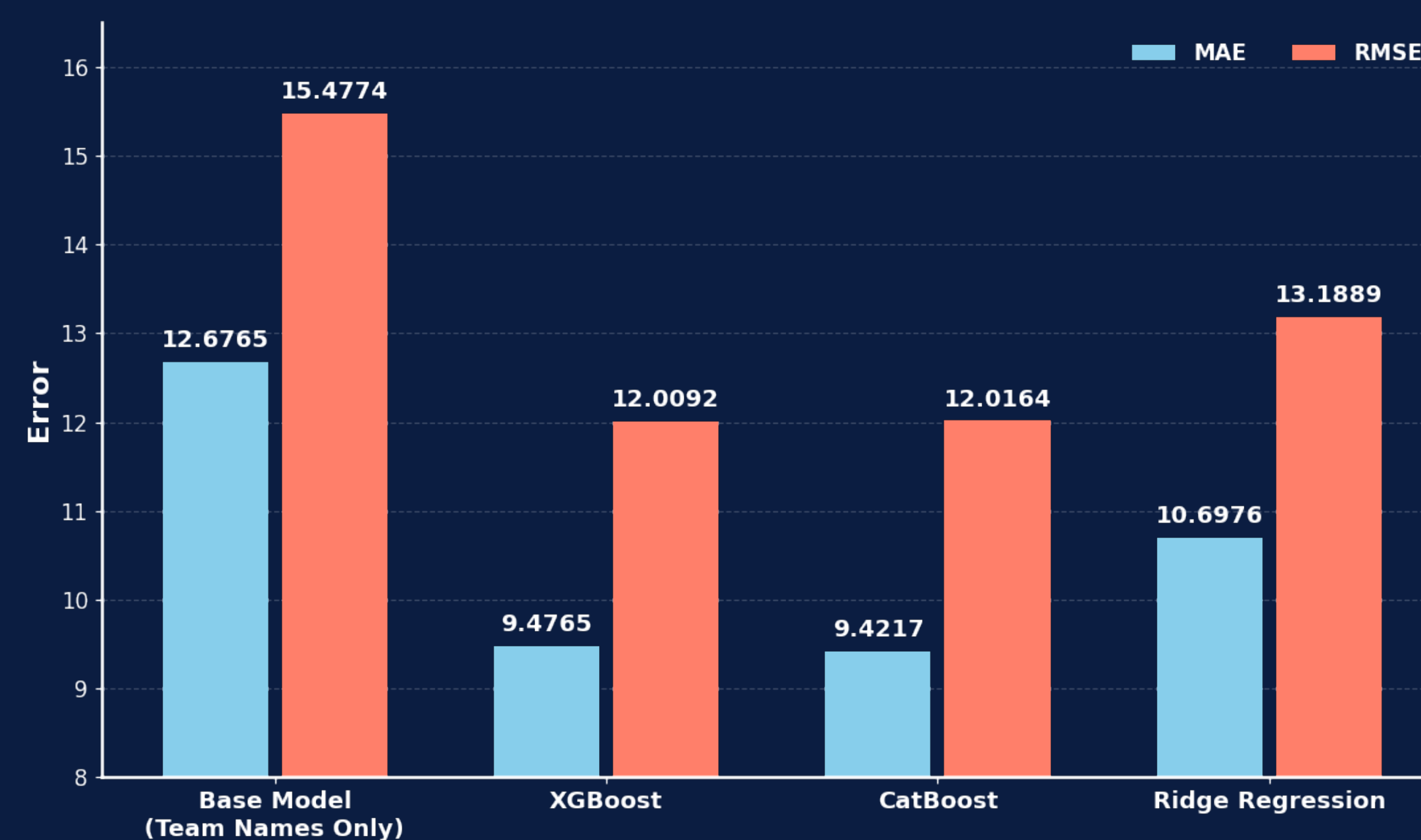
FIGURES

Rolling Window Construction

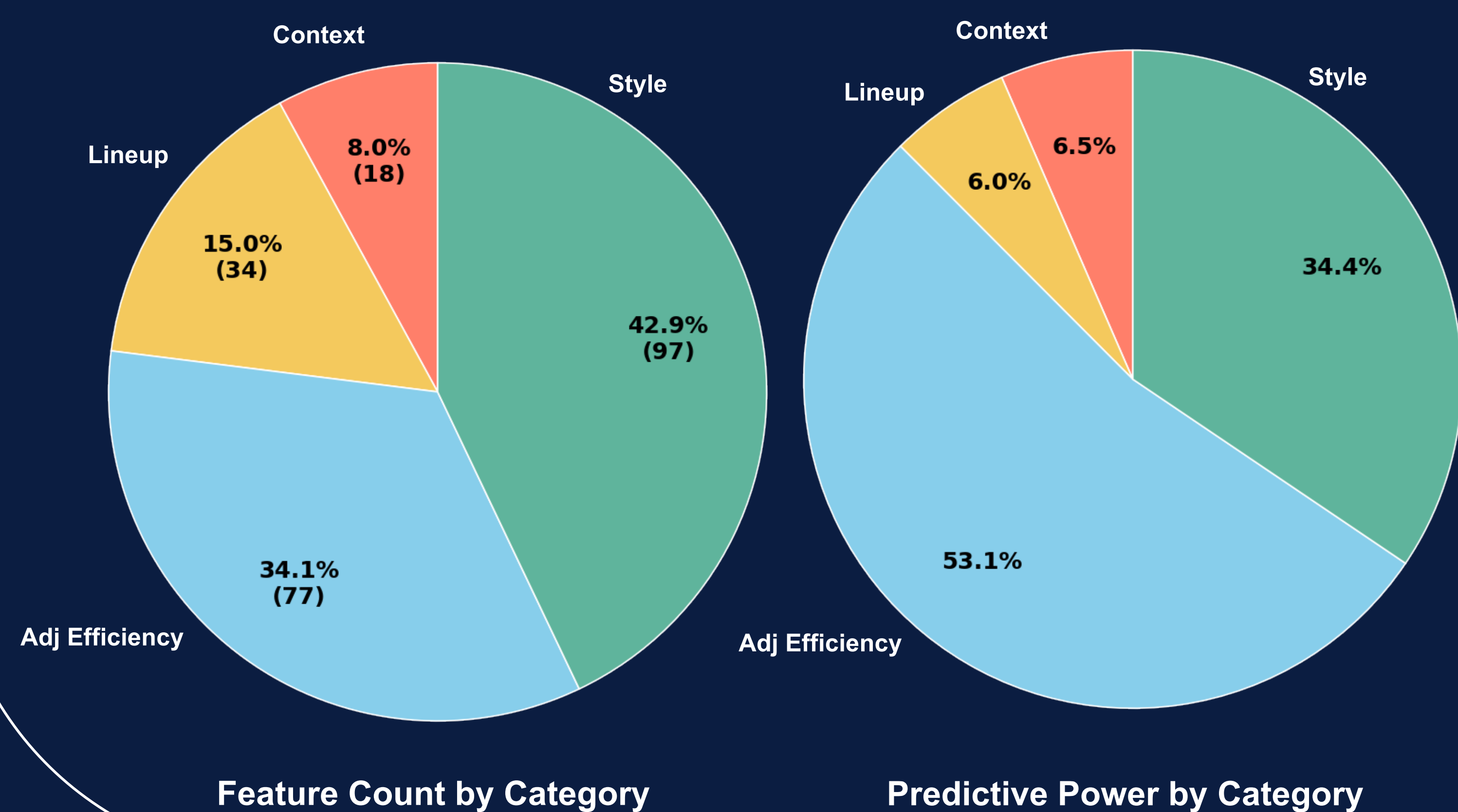


Model testing identified 10 games as the optimal window size

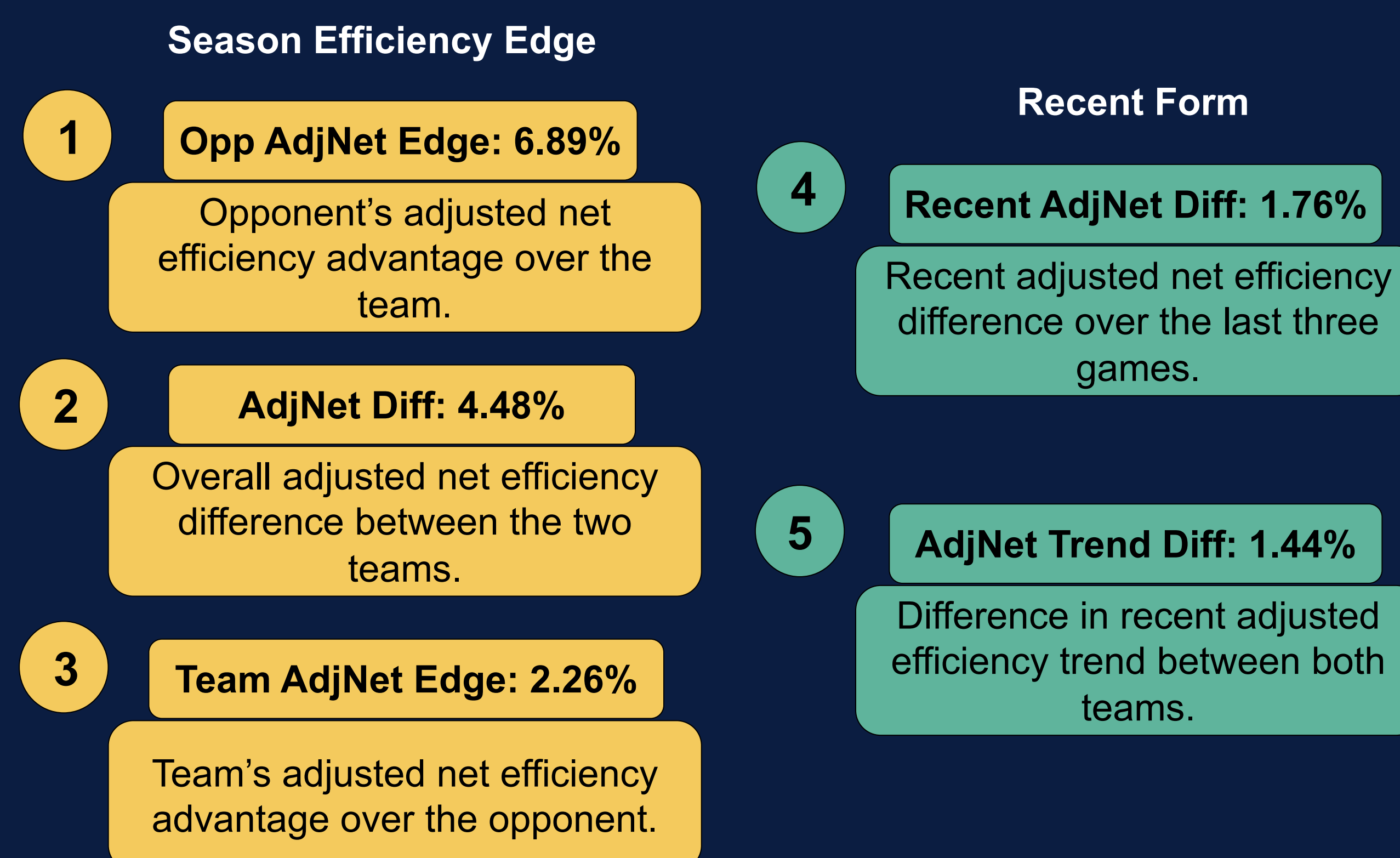
Model Performance Comparison



Training Set Feature Category Breakdown



Top Model Drivers



The strongest predictors are all adjusted-efficiency gap features



RESULTS

Model	MAE	RMSE
Base Model (Team Names Only)	12.6765	15.4774
XGBoost	9.4765	12.0092
CatBoost	9.4217	12.0164
Ridge Regression	10.6976	13.1889

PRACTICAL APPLICATIONS



- More accurate pregame margin forecasts can help coaches and analysts identify which matchups are more dangerous than overall records may suggest.
- Stronger prediction models can support opponent scouting, game preparation, and future lineup decision tools that are better tailored to the specific matchup.



CONCLUSION

The project produced strong predictive results, reducing mean absolute error from 12.68 in the base model to 9.42 in the best model. This improvement shows that the engineered feature set captures meaningful basketball signal, especially through adjusted efficiency. Future gains will likely depend on reengineering style- and lineup-based variables to better reflect the basketball dynamics they were intended to measure.